

Toward Real-time Generation with MiniMax H3

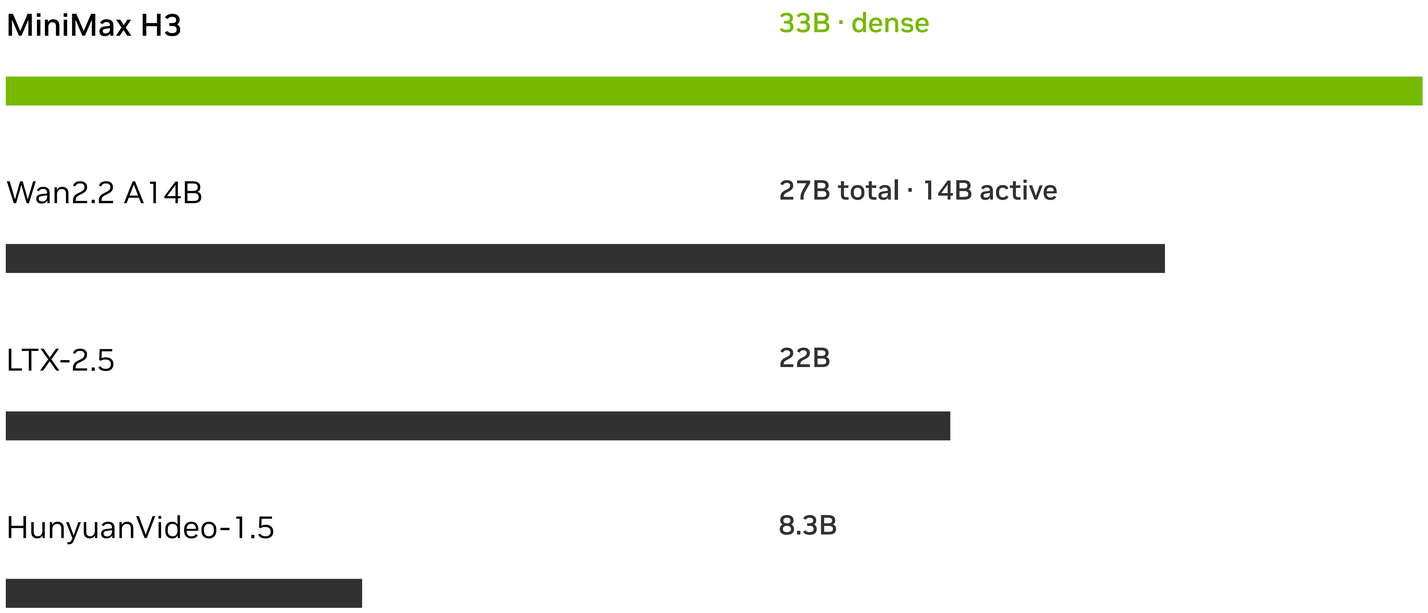
Li Shunyang Devtech Compute NVIDIA

2026-08-23

MiniMax H3 in Context

A large dense backbone for joint audio–video generation.

Generative backbone parameters



Audio + video

Generated together

Multimodal context

Text, image, video and audio

Native 768p

Open H3-Base capability

The challenge: fit the complete audio–video pipeline into a ~5 s budget.

Published full backbone sizes; encoders excluded. Parameter count is not a latency or quality ranking.

NVIDIA

SM120 Cards

The reference system

RTX PRO 5000

×8

SMs	Memory	Power	Interconnect
110	72 GB GDDR7 ECC	300 W	PCIe Gen5 ×16

No NVLink between GPUs. Communication follows the PCIe and NIC topology.

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Baseline: Latency Breakdown

960 × 544 · 5.167 s generated video

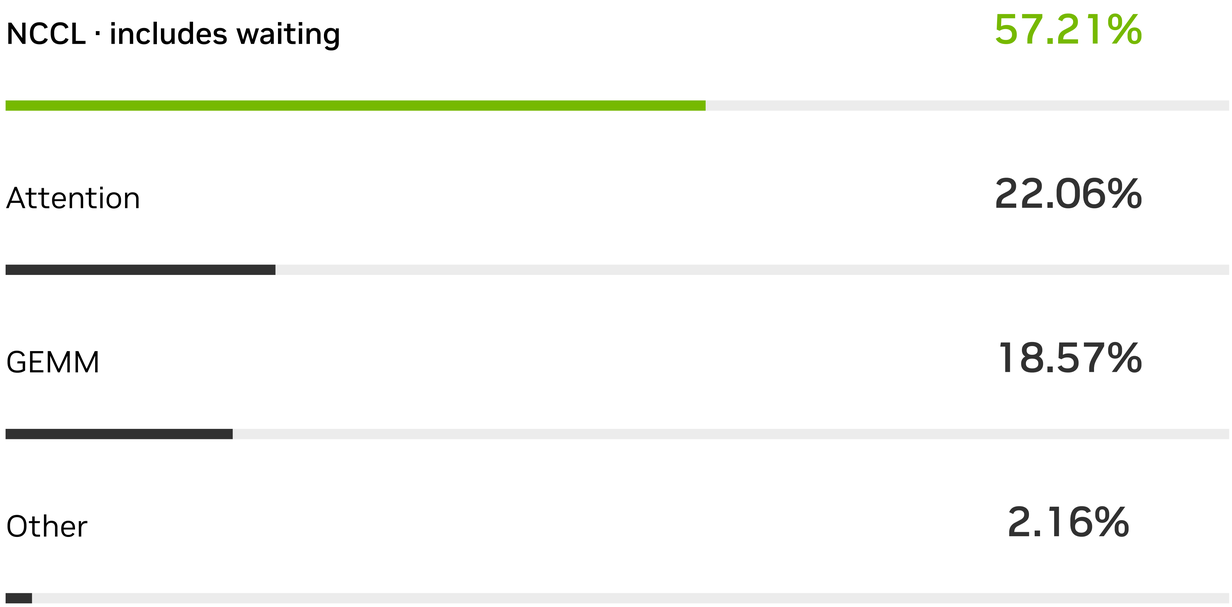
Reference configuration

213.4 s

E2E latency

- 8 × RTX PRO 5000
- 50 steps · BF16 · dense attention
- Ulysses8 · encoder TP8 · VAE tiles8

GPU work by operator



GPU work includes overlap and waiting; these shares are not E2E wall-time components.

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Optimization - Distillation

Reduce how often the model runs.

Base H3

50 model evaluations



Distilled student

8 model evaluations



84%

fewer evaluations

Fewer evaluations reduce compute and communication together.

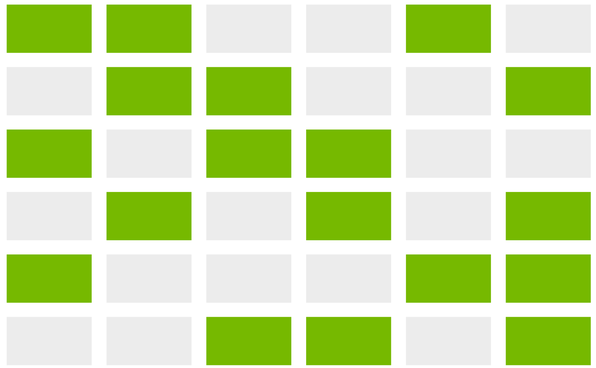
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Optimization - Video Sparse Attention + NVFP4

Do less attention work. Use lower precision where quality allows.

Video Sparse Attention

Attend to selected regions



Content-dependent selection · schematic

NVFP4

Lower precision with QAD

BF16 → NVFP4

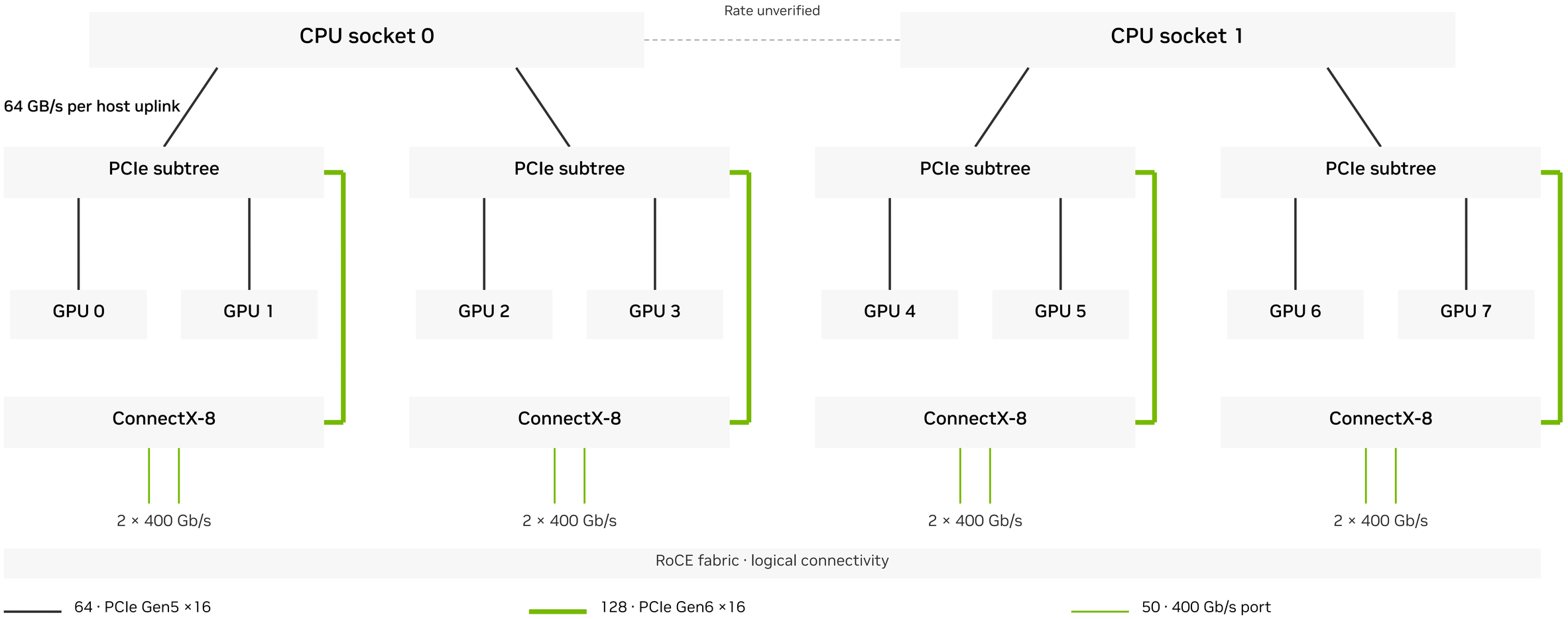
Train for quantization error.
Keep sensitive paths at higher precision.

Evaluate speed and quality together on representative scenes.

NVIDIA

Optimization - Comms

Bandwidth belongs on the topology map.



Line width ∝ nominal one-way GB/s. Raw link rates, not measured payload throughput.

Two GPUs share each host uplink. Choose GPU-affine NICs to avoid that shared route.

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Optimization - Comms

Choose the path, overlap the work, and reduce movement.

Route

Use GPU-affine NICs and RDMA.

Overlap

Transfer the next head band while attention runs.

Streamline

Use direct layouts and replay stable compute.

Overlap transfers with attention



Conceptual head-band schedule

3.1× faster

Grouped E2E: 19.46 s → 6.19 s

Route efficiently. Overlap work. Reduce movement.

Grouped result: RDMA + layout + head overlap + compute graphs.

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From Minutes to Real Time

960 × 544 · 5.167 s output · 8 × RTX PRO 5000

43.4× faster

213.40 s → 4.92 s

Recorded baseline · 50 evaluations

Optimized H3 · 8 evaluations

5.17 s of video generated in 4.92 s · RTF 0.95

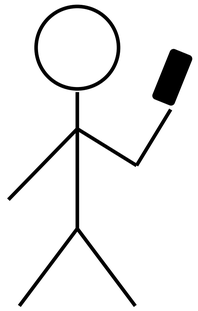
Recorded E2E medians; baseline measured separately. Includes model, schedule and precision changes. Standalone H3.

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From a Question to a Reply

Voice → text → prompt → audio + video

"Hi, could you introduce yourself?"



User

Voice

Qwen3-ASR

0.6B · Speech → text

Qwen3-4B

Reply → H3 prompt

Reference image



H3

Audio + video

"Hi, I'm Miles..."



Avatar reply

PROMPT EXCERPT

... "Hi, I'm Miles. I help with simple, clear answers." ... His lips follow the words.

Hear the avatar's reply

960 × 544 · ~5 s windows · 8 steps · 8 × RTX PRO 5000

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Live Demo

A continuous conversation · final 75% at 3× playback



960 × 544 · ~5 s windows · 8 steps · 8 × RTX PRO 5000

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Takeaways

01

Optimize the communication path.

Topology, bandwidth and less data movement.

02

Validate quantization case by case.

Measure quality and latency together.

03

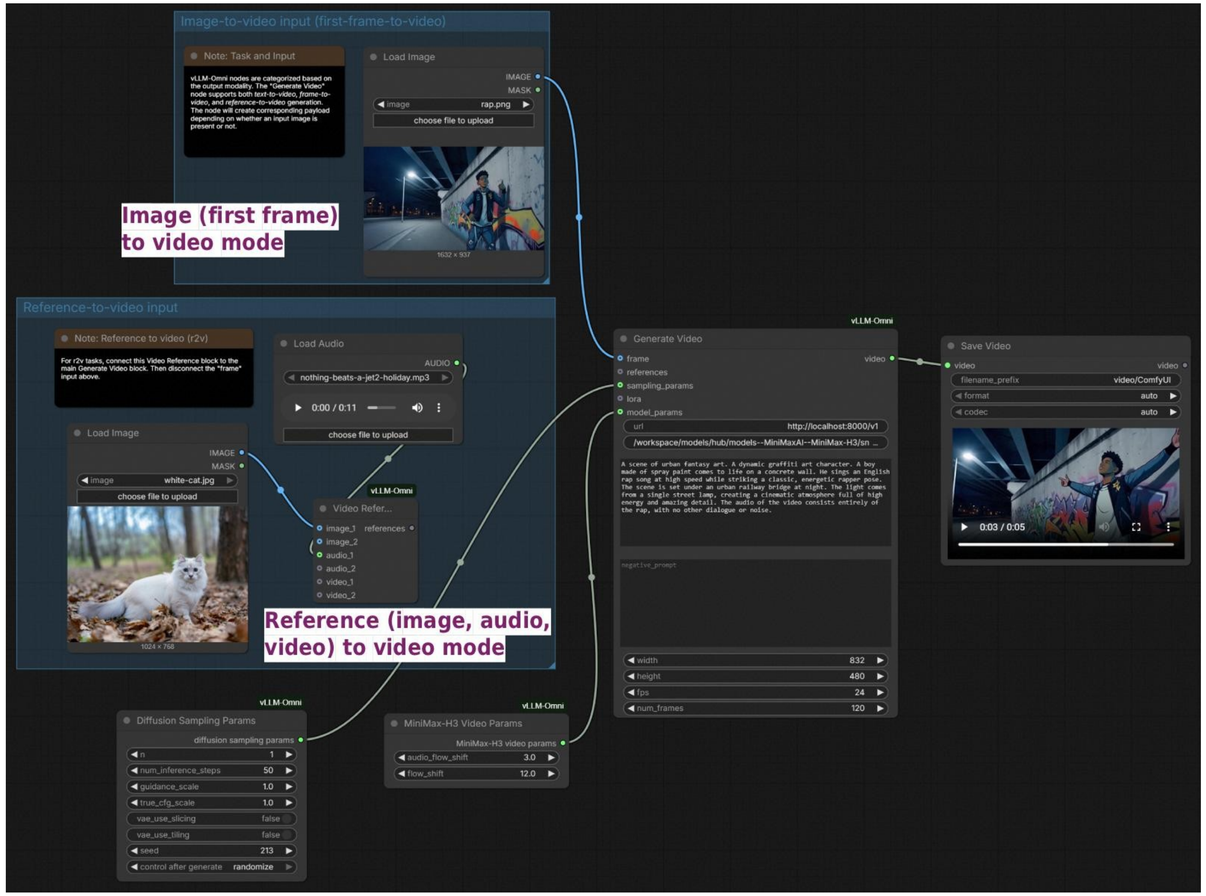
Next: responsive, richer interaction.

Autoregressive diffusion + prompt enhancement.

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vLLM-Omni for Creators — ComfyUI

Visual workflows, powered by a local or remote model service.



Compose in ComfyUI. Run models in vLLM-Omni.

AVAILABLE ON MAIN

Compose visually

Image, video and audio workflows.

Serve separately

Models run in vLLM-Omni.

Bring H3 into the graph

Text, keyframes or reference media.

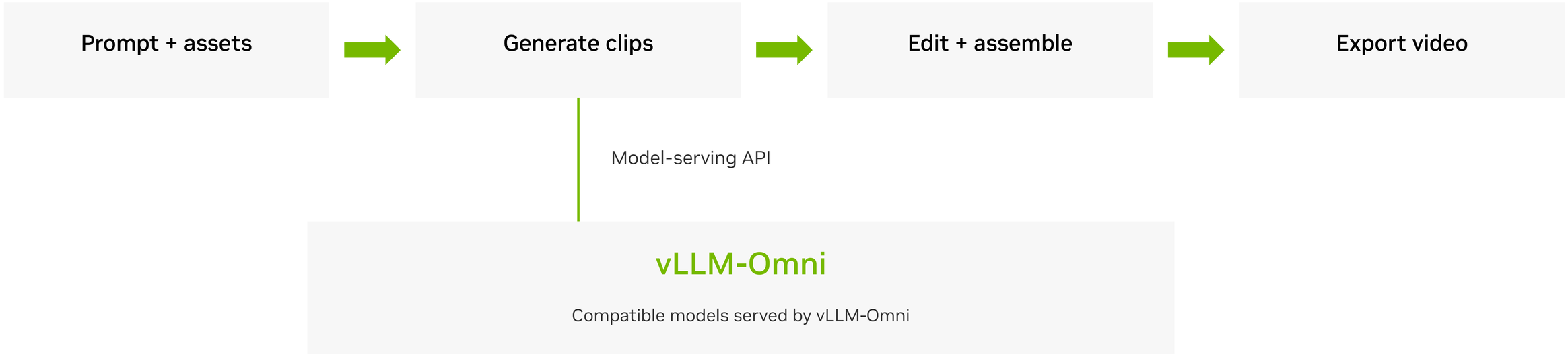
Official UI example, not benchmark settings. Upstream main verified 30 Oct 2026 · 08177216

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vLLM-Omni for Creators — Video Studio

Planned open-source release

A developer tool for video generation and editing.



One workflow, from generated clips to an edited video.

Video Studio is a working title. Diagram shows the planned workflow and backend integration.

vLLM-Omni Networking Hour!

https://github.com/vllm-project/vllm-omni

https://blog.vllm.ai/2025/11/30/vllm-omni.html

https://communityinviter.com/apps/vllm-dev/join-vllm-developers-slack

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